



QGIS 4.x PLUGIN / VERSION 0.4.1

MILEVA

User guide

Mapping-Informed Landscape Ecotopes for Versatile Acoustic Sampling

Purpose. Build landscape-informed sampling designs, generate a complete grid and a reproducible random baseline, summarize a raster proxy around proposed or existing recorder sites, and compare each design with the complete reference.

Scientific origin. This plugin implements selected cartographic and distribution-comparison stages of the workflow in Martínez-Arias et al. (2026), with the operational sequence informed by Supplementary Information S10. The article evaluates sampling designs across three Colombian ecosystems using acoustic indices and interpolated soundscape surfaces. MILEVA is a planning aid and does not reproduce those acoustic analyses.

Key distinction: the complete scenario is an idealized comparison baseline, not a recommendation to deploy a recorder at every grid point. The default 200 m is a spatial design scale, not an estimate of biological detection or microphone range.

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1 Before you begin

Install the plugin ZIP through **Plugins → Manage and Install Plugins → Install from ZIP**. In the Processing Toolbox, open **MILEVA → Workflow steps**. Enable the **GRASS GIS Processing Provider** for watershed tessellation.

Input	What to check
Landscape proxy	A continuous raster of the landscape variable you want the design to represent; use band 1. A spectral index or a fused environmental surface can serve as the proxy.
Study area	A polygon layer, a valid-data raster, or both for the complete grid. If both are supplied, point centers must meet both masks.
Coordinates	Use a projected CRS with metre units for distance and area steps. Match CRS between vector/raster inputs where a tool requires it.
Recorder sites	Optional existing points for selecting real candidate sites from tiles. Reproject them to the tile CRS beforehand.

2 Rescale landscape proxy

Run **STEP II - Rescale landscape proxy (1-100)**. Choose a continuous input raster; the default new minimum and maximum are 1 and 100. The rescaled output is the input to the watershed step. NoData is preserved by the underlying QGIS raster rescaling operation.

3 Create landscape tiles

Run **STEP III (a) - Tessellation: basins / half-basins**. Select the rescaled raster, set **Recorder radius (meters)** (default 200), and choose **Basins** or **Half-basins**. MILEVA converts $\pi \times \text{radius}^2$ to an approximate pixel threshold, runs GRASS r.watershed, and polygonizes the selected output. Temporary layers are named **Basins** or **Halfbasins**. This treats the proxy as a segmentation surface; it does not require an elevation model.

SLIC: MILEVA does not run SLIC. If you have SLIC polygons produced externally (for example, following the S10 notebook), load them in QGIS and use them as tiles in Step IV. Land-cover polygons can be used in the same way.

4 Build a complete and random scenario

Run **STEP III/IV - Complete grid + random subsample**. Select study-area polygons, a study-area raster, or both. MILEVA places a regular grid inside polygon geometry and/or over raster cells with valid band-1 values. It excludes raster NoData and, by default, the additional value 999.

Control	Effect
Minimum distance / complete grid spacing	Default 200 m. Centers are placed on one regular grid; any two grid points in either output are at least this far apart.
Random sample size	A positive count selects that many points. Set 0 to use the percentage instead.
Random sample percentage	Default 20% of the complete candidates; at least one point is selected. The count takes precedence when positive.
Random seed	Default 12345. Reusing the same inputs, grid spacing, and seed reproduces the random subset.
Outputs	Complete and Random point layers, each with sample_id . Random is drawn from Complete.

Geometry of the baseline: this is a random subset of fixed grid sites. The tool does not scatter continuously random coordinates across the polygon. Choose a meaningful spatial extent and inspect the grid before interpreting any comparison.

5 Select points from tiles

Run **STEP IV - Select sampling points** with Basins, Halfbasins, or any polygon tessellation. The algorithm finds the pole of inaccessibility inside each tile. If you supply **Existing recorder points**, it selects the nearest existing recorder to each pole and avoids selecting the same recorder twice; otherwise, the poles are proposed sites. **Pole tolerance** controls the pole calculation in map units (default 1).

The point output has **tile_id**, **recorder_id** (the input feature ID when a recorder is selected), and **distance_m** from the pole. That distance is in metres only for a metre-based CRS. A tile may not contribute a unique recorder if its nearest candidate was already selected.

6 Summarize the proxy around each design

Run **STEP V - Buffer + zonal proxy mean** separately for Complete, Random, Basins, Halfbasins, and any other selected-point design. Use the **same landscape proxy raster and buffer radius** for a fair comparison. Enter a distinct **Origin** label for each design, for example **complete, Random, Basins**, or **Halfbasins**.

Each point is buffered by the selected radius (default 200 m). QGIS calculates the mean of valid band-1 raster cells in that buffer. The polygon output contains **L_proxy_mean** and **Origin**. A buffer with no valid raster values may have a missing mean; the comparison step skips non-numeric values. Points and raster must share the same projected CRS.

7 Compare separate output layers

Run **STEP VI - Statistical comparison (KS + Wasserstein)**. Select the **Complete scenario layer (Step V output)**, then choose one or more **Subsample layers (Step V outputs)**. Leave **Already merged layer** empty. The numeric field defaults to **L_proxy_mean** and must have the same name in every layer. No manual merge is needed.

Mode	Required inputs
Separate layers (recommended)	A complete Step V layer and at least one other Step V layer. Origin labels or layer names identify methods.
Legacy merged layer	A single existing merged layer with a numeric value field, a group field (default Origin), and a reference group (default complete). Leave separate inputs empty.

Practical sequence: Complete points → Step V with Origin=complete → reference layer. Random points → Step V with Origin=Random → one comparison layer. Basins tiles → Step IV → Step V with Origin=Basins → another comparison layer. Then compare the separate Step V layers in one Step VI run.

A comparison row is created for each subsample layer. When a layer has one distinct Origin value, it becomes **Method**; otherwise MILEVA uses the layer name. Repeated labels receive a numeric suffix.

8 Read the comparison table

Column	Interpretation
Method	The subsample design being compared with the complete reference.
N_ref / N_sub	Counts of valid numeric proxy means in the reference and subsample. Compare these with the planned recorder effort.
KS_statistic	Largest absolute gap between the two empirical cumulative distributions. 0 means identical empirical distributions; smaller is more similar.
KS_p_value	Approximate two-sample Kolmogorov-Smirnov p-value. Small values provide evidence against a shared distribution. A large p-value does not prove equivalence.
Wasserstein	One-dimensional Wasserstein-1 distance between distributions of mean proxy values. Smaller is closer, in the same units as L_proxy_mean (usually rescaled proxy units).
S10_sorted_delta	Mean absolute difference between the first min(N_ref,N_sub) sorted values in the two samples. Retained to compare with the S10 notebook; generally differs from true Wasserstein when sizes differ.

Illustrative reading: if Basins and Random use similar **N_sub**, and Basins has lower **KS_statistic** and **Wasserstein**, its sampled proxy means more closely resemble the complete proxy distribution. This supports Basins for that proxy and scale; it does not establish better acoustic representativeness without sound data.

Statistical cautions: KS p-values are asymptotic and can be unreliable with few samples or many tied proxy values. Spatially nearby buffers can overlap, so observations may be dependent. Favor effect sizes (KS statistic and Wasserstein), sample counts, maps, and field constraints over a single p-value.

9 Make and document a decision

- Inspect each tessellation and point layer: coverage, edge behavior, inaccessible sites, duplicate recorder selection, and spacing.
- Check that Complete and every subsample use the same proxy, band, rescaling, buffer radius, CRS, and study area.
- Read **N_sub** with KS and Wasserstein to balance resemblance against the number and cost of recorders. Compare multiple random seeds if chance variability matters.
- Record the proxy source, CRS, radius, seed, grid spacing, chosen method, output layer files, and field feasibility decisions.

Scope and relationship to the publication

MILEVA provides spatial preparation and a distributional comparison of **landscape proxy means**. It does not calculate acoustic indices, interpolate soundscape surfaces with kriging, estimate spatial prediction error, or choose a final field design automatically. The article evaluates those additional acoustic and spatial performance questions. Its reported relative performance among strategies should not be assumed to transfer unchanged to a new proxy, habitat, taxon, or sampling scale.

The article used random percentages of 20%, 40%, 60%, and 80% with repeated draws. This plugin creates one reproducible random draw per run; change the seed and repeat runs when you need a distribution of random-baseline outcomes. The 200 m default reflects the paper's spatial independence design choice and may be changed for the study objective.

Troubleshooting

Message / symptom	Check
GRASS r.watershed is unavailable	Enable the GRASS GIS Processing Provider in QGIS, then rerun Step III.
Distance or CRS error	Reproject input layers to one projected, metre-based CRS. Verify raster and vector CRS match in Steps III-V.
No grid points / too many candidates	Review the polygon mask and raster valid values; adjust spacing or clip the area.
No valid values in comparison	Confirm each Step V output has numeric L_proxy_mean values, including the complete reference.

Citation and credits

Martínez-Arias, V. M., Paniagua-Villada, C., Guerrero, M. J., & Daza, J. M. (2026). A workflow to optimize spatial sampling in ecoacoustic studies. *Landscape Ecology*, 41, article 126. <https://doi.org/10.1007/s10980-026-02372-5>. See Supplementary Information S10 for the original workflow.

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